HEMANTHA KRISHNA CHALLA

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EDUCATION

The University of Texas at Dallas *Master of Science, Computer Science (Cybersecurity)* Graduation: May 2026

- **NSA Designations:** Cyber Operations (CAE-CO) & Cyber Defense (CAE-CD) ***GPA 3.70***

SRMAP University, AP, India *Bachelor of Technology in* Computer Science (Cybersecurity) Graduated: May 2024

TECHNICAL SKILLS

- **Offensive Security**: OSINT, External/Internal Penetration Testing, Active Directory Exploitation (Kerberoasting, Pass-the-Hash, BloodHound), Network Pivoting, and Technical Report Writing.
- **Incident Response:** SIEM (Splunk, Elastic), Snort IDS, Wireshark, Phishing Analysis, EDR, Digital Forensics
- **Programming Languages:** Python, C, C++, Java, bash, x86 Assembly
- **Reverse Engineering & IoT:** Ghidra, Binwalk, Firmware Extraction, UART/I2C Hardware Analysis, GDB
- **Network & Infrastructure:** Active Directory, Network Enumeration, Wireless Security, Linux, Windows

CERTIFICATIONS

- Practical Network Penetration Tester (PNPT)
- Practical Junior Penetration Tester (PJPT)
- Practical IoT Pentest Associate (PIPA)
- CompTIA Security+ (SY0-701)
- Practical SOC Analyst Associate (PSAA)

PROJECT EXPERIENCE

Practical Network Penetration Tester (PNPT) | TCM Security

- External Pentest & OSINT: Conducted comprehensive open-source intelligence gathering to identify leaked credentials and external attack vectors, simulating a real-world adversary.
- Network Pivoting & AD Exploitation: Leveraged external footholds to pivot into a multi-forest Active Directory environment, executing lateral movement and privilege escalation to achieve Enterprise Admin privileges.
- Delivered a professional report detailing critical attack chains and defining strategic remediations.
- Live Debrief: Defended findings and remediation strategies in a live, one-on-one debrief with a security expert.

Toyota HackFesta 2025 - 1st Place Winner (Team)

- Led a high-performance team to **1st place** in a two-day capture-the-flag event by identifying and exploiting vulnerabilities in simulated automotive CAN and Ethernet networks.
- Mitigated critical automotive security threats in a simulated environment, demonstrating practical application of CAN and Ethernet network defense strategies required by Toyota's Product Cybersecurity Group.

Capture The Flag (CTFs) – HackTheBox and TryHackMe | Top 1% Globally

- Achieved a global rank of 9,196 by mastering 262+ hands-on labs focused on Red Team operations, Windows/Linux privilege escalation, and Active Directory exploitation, Maintaining a 482+day learning streak on tryhackme.
- Published 40+ technical walkthroughs documenting adversarial methodologies and exploit chains.

Security Operations & Incident Response – Practical SOC Analyst Associate (PSAA) | TCM Security

- Executed full-cycle incident response for simulated network intrusions, identifying root causes of breaches.
- Authored professional remediation reports detailing security gaps and actionable defense recommendations.

Practical IoT Pentest Associate (PIPA) -TCM Security

- Identified 15+ vulnerabilities in embedded Linux firmware, including a Critical Remote Code Execution (RCE) and privilege escalation flaws, by applying advanced reverse engineering techniques.
- Compromised encryption via hardware analysis (UART, I2C) and Ghidra, simulating vehicle hardware attacks.

PUBLICATIONS

IEEE ASSIC 2024: Task Offloading Technique Selection in Mobile Edge Computing (MEC)

Journal of Technology: Machine Learning Intrusion Detection with Explainable AI

WORK EXPERIENCE

CS Grader – Undergraduate Aug 2025 - Present

- Evaluate assignments and projects, providing feedback to students on discrete mathematics.

Web Development Intern at Jayaho, CCC Digital India Pvt. Ltd., Hyderabad, Telangana, India Jun 2022 – Aug 2022

- Applied secure coding practices (OWASP Top 10) to enhance website resilience, increasing engagement by 20%.